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Einstein's theory confirmed

Research has confirmed Einstein's theory that Earth is a free-falling elevator within the Sun's gravity <https://digit.in/jul18-79-2>

healed better using this process too. What remains is the standardisation of the method.

Another technique that has shown promise is the use of biomaterials and tissue engineering. The use of amniotic membranes or collagen sheets not only speeds up healing from burn wounds, it also reduces things like itching and pain in the duration of the healing. Even the scars are less visible. Microbial cellulose and electrospun nanofibrous dressings are other alternative materials being used to achieve similar results. At this point, researchers have decided to take it one step further by incorporating drugs and bioactive compounds to accelerate it even further.

One of the leading combinations in this area is ginsenoside Rg3-loaded electrospun poly (lactic-co-glycolic acid) fibrous membranes (yeah, that is a mouthful). Even Norfloxacin-loaded collagen seems to be effective in reducing scarring and increasing cell growth for wound closure. Stem cell technology and bioengineered skin substitutes have shown better rates of healing than the natural process. Despite that, none of this is really the kind of accelerated healing we have seen in fiction - it is not even close, either in speed or in capabilities to regenerate without external influence. What would it take for us to achieve that? The key, quite interestingly, could lie with a certain Merc with a Mouth and the work of a certain 19th century fictional doctor.

TRUE REGENERATION

If you do know about Deadpool, you must be familiar with his regenerative powers that landed him on the subtitle for this article. Deadpool can regrow entire limbs overnight - and his ability has been found to be scientifically accurate! The human body takes about 18 years to reach its full size and even nature's super healers, the salamanders, take about a few months to regrow a lost body part. So how does Deadpool do it? By temporarily stopping what stops cells from growing

uncontrollably - yes, the body has to tell itself to stop growing.

Without proto-oncogenes and tumor suppressor genes in the human body, cells would, and do, grow without check, leading to conditions like cancer. According to stem cell biologists, tumor suppressor genes can also selectively deactivate genes to promote tissue regeneration. It is these genes and oncogenes that Deadpool has immense control over, and that we need to harness for true regeneration. Essentially, Deadpool can talk to his genes. Yes, we did just tell you that you need a superpower - and before you disregard this article for being too speculative, let us tell you that another fictional character, and the work of a certain scientist, shows us exactly how to get that superpower.



Michael Levin, riding on the shoulders of scientists from centuries ago, could unlock the key to true, rapid regeneration

ELECTRIFYING

Unlike Deadpool, not many know about the story of Frankenstein's creature (yes, the creature was not called Frankenstein, the man who created him was). TLDR, after putting him together, Dr. Frankenstein essentially used a surge of electricity and a combination of certain chemicals to bring him to life. Michael Levin's work could be described in the same vein.

Working out of his lab at Tufts University, Michael Levin has been

working on establishing a link between bioelectricity and the knowledge that our cells possess about how to grow - and possibly regrow - parts of our body. And he's achieved quite a degree of success with that. His work involved using cell wall pumps and other methods on tadpoles to 'make the body believe' that it can regrow a lost tail - and it did. It has led to its fair share of Frankenstein's monsters as well - misplaced organs in frog embryos, eyes on tadpole tails, flatworms with "cat-like" heads etc. If you have a feeling that this is never going to work on complex organisms like humans, let us inform you that in 2016, Tufts University created the Paul G. Allen discovery centre and appointed him as the head, an honor that came with a \$10 million grant

and the possibility of \$10 million more if the work shows promise after four years.

The field of accelerated, explicitly-controlled healing might be entirely resting on the work of this one person looking to create his own niche mutation of computing and genetics. As some of his contemporaries have put it, Michael Levin could someday get the Nobel Prize, or put the question of bioelectric coding, and the possibilities it holds, to rest once and for all. 